

QA CHECKLIST

Software Quality Verification – Pre-Release Checklist

Document Type	QA Checklist
Version	1.0
Application	Student Registration System
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Organisation	The Career Insights Hub LLC
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Build Version	Build-1.0
Environment	QA / Staging
Overall Status	In Review

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1. Purpose & Scope

This QA Checklist is a structured pre-release verification document designed to ensure the Student Registration System meets defined quality standards before being promoted to production. It covers all critical checkpoints across functionality, input validation, usability, security, performance, and overall quality standards. Each item is individually reviewed and marked with a result status.

Status Key:

PASS	FAIL	WARN	N/A
Checkpoint passed — meets requirement	Checkpoint failed — defect logged	Partial pass — risk noted, monitor	Not applicable to this build

2. Functionality Checklist

2.1 Student Signup

ID	Checkpoint / Criteria	Result	Notes / Observations
FC-001	New student can register with all valid required fields	PASS	Registration successful — redirects to login page
FC-002	Duplicate email registration is rejected with an error message	FAIL	Duplicate email accepted — duplicate account created (REF: BUG-003)
FC-003	Confirmation/welcome email sent after successful registration	PASS	Email received within 60 seconds of signup
FC-004	All mandatory fields display validation errors when left blank	PASS	All required field errors trigger correctly on submit
FC-005	Password and confirm password mismatch is rejected	PASS	Error: 'Passwords do not match' displayed correctly

2.2 Login

ID	Checkpoint / Criteria	Result	Notes / Observations
FC-006	Student login with valid credentials redirects to student dashboard	PASS	Login successful — correct dashboard displayed
FC-007	Admin login with valid credentials redirects to admin dashboard	PASS	Admin dashboard loaded with correct role
FC-008	Login with incorrect password displays error — no session created	FAIL	Login accepted with wrong password (REF: BUG-001)
FC-009	Account lockout triggered after 5 failed login attempts	FAIL	No lockout observed — unlimited attempts allowed (REF: BUG-002)

ID	Checkpoint / Criteria	Result	Notes / Observations
FC-010	Logout destroys session — back button does not restore dashboard	FAIL	Session persists after logout via browser back button (REF: BUG-006)
FC-011	Unauthenticated users redirected to login page on protected routes	PASS	Redirect to /login confirmed for /dashboard and /profile

2.3 Course Enrollment

ID	Checkpoint / Criteria	Result	Notes / Observations
FC-012	Logged-in student can enroll in an available course	PASS	Enrollment confirmed — course appears in My Courses
FC-013	Duplicate enrollment in same course is blocked with error	FAIL	Duplicate enrollment accepted — course appears twice (REF: BUG-005)
FC-014	Enrollment in a full course is blocked — seat count does not go negative	FAIL	Student enrolled in full course — seat count became -1 (REF: BUG-009)
FC-015	Student can unenroll from an enrolled course	PASS	Course removed from profile — seat count incremented correctly
FC-016	Course catalog search and filter returns accurate results	PASS	Search by keyword and department filter working as expected

2.4 Profile Update

ID	Checkpoint / Criteria	Result	Notes / Observations
FC-017	Student can update phone number and address — changes saved correctly	PASS	Profile updated — new values displayed immediately
FC-018	Oversized profile photo upload is rejected with error message	FAIL	8MB image accepted — no error shown (REF: BUG-010)
FC-019	Email field is read-only and cannot be changed without admin approval	PASS	Email field locked — no edit possible from student profile
FC-020	Required Last Name field cannot be cleared and saved	FAIL	Empty Last Name saved with no error (REF: BUG-008)

2.5 Admin Dashboard

ID	Checkpoint / Criteria	Result	Notes / Observations
FC-021	Admin can view full registered student list	PASS	All students listed with correct details
FC-022	Admin can search and filter students by name and ID	PASS	Search returns accurate, filtered results
FC-023	Admin can deactivate a student account	PASS	Student deactivated — login blocked for that account
FC-024	Admin can add a new course with all details	PASS	Course saved and visible in student catalog
FC-025	Admin can edit course capacity and details	PASS	Changes saved and reflected in catalog immediately
FC-026	Students cannot access Admin Dashboard via direct URL	FAIL	Student role accessed /admin/dashboard directly (REF: BUG-007)

3. Validation Checklist

3.1 Input Field Validation

ID	Checkpoint / Criteria	Result	Notes / Observations
VC-001	Email field rejects invalid formats (missing @, domain, etc.)	PASS	Inline error displayed for malformed email formats
VC-002	Password complexity rule enforced — min 8 chars, uppercase, number, special	FAIL	Weak passwords like '123456' accepted with no error (REF: BUG-004)
VC-003	Phone number field rejects non-numeric characters	PASS	Letters in phone field trigger validation error
VC-004	Text fields enforce maximum character length limits	WARN	255-char limit works for name; some fields have no visible max limit set
VC-005	File upload field rejects non-image file types (.pdf, .exe)	WARN	Only image size tested; file type validation not fully verified
VC-006	All required fields marked with asterisk (*) or indicator	PASS	All mandatory fields visually marked in the UI
VC-007	Form cannot be submitted via direct API call bypassing UI validation	FAIL	API accepts invalid data when frontend validation is bypassed — server-side validation missing
VC-008	SQL injection attempt in login fields is blocked	PASS	Input sanitised — no SQL execution, error returned cleanly
VC-009	XSS attempt in name fields is sanitised and not rendered	PASS	HTML/script tags stripped from output — no XSS execution
VC-010	Boundary values tested for enrollment capacity (0, 1, max)	FAIL	Zero-seat course still accepts enrollment (REF: BUG-009)

3.2 Error Message Validation

ID	Checkpoint / Criteria	Result	Notes / Observations
VC-011	Error messages are clear, descriptive, and user-friendly	PASS	Messages describe the issue and guide user — no technical stack traces
VC-012	Error messages do not expose internal system details or stack traces	PASS	Generic error messages confirmed — no DB or server info leaked
VC-013	Success messages confirm completed actions clearly	PASS	Enrollment, profile save, and logout all show confirmation messages
VC-014	Form errors highlight the specific field that failed validation	PASS	Fields outlined in red with inline error text on validation failure

4. Usability Checklist

4.1 User Interface & Navigation

ID	Checkpoint / Criteria	Result	Notes / Observations
UC-001	Application loads within 3 seconds on standard connection	PASS	Average page load: 1.8 seconds on broadband
UC-002	Navigation menu is consistent and accessible from all pages	PASS	Header nav present on all pages — links work correctly
UC-003	All buttons and links are clearly labelled and functional	PASS	No broken links or unlabelled buttons found
UC-004	Page layout is consistent across all 5 modules	PASS	Consistent header, footer, and content structure throughout
UC-005	Tab order on forms follows a logical top-to-bottom flow	PASS	Keyboard tab order matches visual field order on signup and login
UC-006	User receives feedback on all actions (save, enroll, logout)	PASS	Confirmation messages shown for all key user actions
UC-007	Application is responsive on desktop browsers (Chrome, Firefox, Edge)	WARN	Tested on Chrome 120 — Firefox and Edge not yet verified
UC-008	Back button behaviour is consistent and does not break navigation	FAIL	Back button restores session after logout (REF: BUG-006)
UC-009	404 and error pages are styled and provide navigation back to home	WARN	404 page exists but lacks navigation back to dashboard
UC-010	Loading indicators displayed during long operations	WARN	No spinner or loader shown during course enrollment submit

4.2 Accessibility & Standards

ID	Checkpoint / Criteria	Result	Notes / Observations
UC-011	Form labels are properly associated with their input fields	PASS	Labels linked via 'for' attribute — screen reader compatible
UC-012	Images and icons include alt text or ARIA labels	WARN	Profile photo upload icon missing alt text
UC-013	Colour contrast meets WCAG 2.1 AA minimum standards	PASS	Contrast ratios checked — all body text passes 4.5:1 minimum
UC-014	Application is operable using keyboard only (no mouse required)	PASS	All core actions completable via keyboard tab and enter

5. Overall Quality Standards

5.1 Security

ID	Checkpoint / Criteria	Result	Notes / Observations
QS-001	HTTPS enforced across all pages — no mixed content	PASS	All pages served over HTTPS — no HTTP downgrade observed
QS-002	Session tokens are secure, random, and expire after inactivity	WARN	Session expiry configured but session-after-logout not invalidated server-side
QS-003	Role-based access control enforced at server level — not just UI	FAIL	Student accessed admin route via direct URL (REF: BUG-007)
QS-004	Passwords stored as hashed values — not plain text	PASS	bcrypt hash confirmed in DB — no plaintext storage
QS-005	CSRF protection implemented on all state-changing forms	WARN	CSRF token present on signup/login — not verified on all API endpoints

5.2 Performance

ID	Checkpoint / Criteria	Result	Notes / Observations
QS-006	Login page responds within 2 seconds for single user	PASS	Response time: 0.6 seconds — within SLA
QS-007	Course enrollment responds within 2 seconds for single user	PASS	Response time: 0.9 seconds — within SLA
QS-008	System handles 50 concurrent users without errors	WARN	Not yet load tested — JMeter performance test pending pre-release
QS-009	Database queries execute without timeouts under normal load	PASS	No query timeouts observed during manual testing session

5.3 Documentation & Code Quality

ID	Checkpoint / Criteria	Result	Notes / Observations
QS-010	Test cases mapped to requirements in RTM — full traceability	PASS	RTM covers all 36 requirements to 42 test cases (Assignment 2)
QS-011	All identified defects logged with full details in JIRA/tracker	PASS	10 defects logged with steps, severity, priority, and status
QS-012	Browser compatibility matrix defined and executed	WARN	Chrome 120 tested — Firefox and Edge matrix pending
QS-013	Release notes document provided for this build	WARN	Release notes provided but do not include list of known open defects
QS-014	QA sign-off checklist completed and approved before release	FAIL	6 open Critical/High defects remain — sign-off blocked

6. Checklist Summary

Category	Total Items	PASS	FAIL	WARN	N/A
Functionality	21	13	7	0	1
Validation	14	7	4	3	0
Usability	14	7	1	5	1
Overall Quality	14	6	2	5	1
TOTAL	63	33	14	13	3

Release Recommendation: Application is NOT READY FOR RELEASE in its current state. 6 Critical/High defects remain open and must be resolved and verified before QA sign-off can be granted. All FAIL items must be addressed and retested.

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